

Clustering

The K-means Algorithm, Its Objective, Initialisation, and Choosing K

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Week 7 · Part 1

Module 3: learning without labels

Modules 1 and 2

Every training example came with an answer: $(\mathbf{x}^{(i)}, y^{(i)})$. The model learned to map $\mathbf{x}$ to y — **supervised** learning.

Module 3

The data are only $\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(m)}$. There is no y to predict. The task is to find *structure*: groups, unusual points, hidden factors — **unsupervised** learning (Week 1's definition).

Today

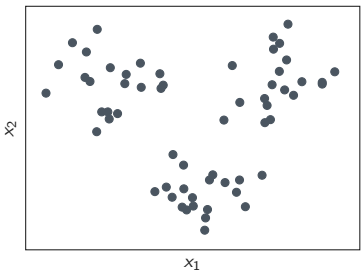
1. What is clustering?
2. K-means, watched step by step
3. The algorithm, by hand
4. The optimisation objective
5. Initialising K-means
6. Choosing the number of clusters

Part 2: anomaly detection, the other unsupervised workhorse of this week. Several frames are animations: advance them one click at a time.

What Is Clustering?

Labels or no labels

the same 60 points, with no labels



Clustering

Given unlabelled data, find groups — **clusters** — of points that are similar to each other and different from points in other groups.

Where it is used

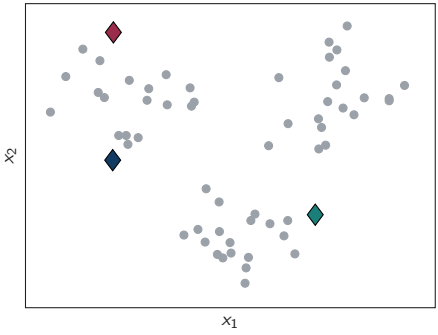
- Grouping similar news articles.
- Market segmentation: customers who buy alike.
- DNA analysis; grouping stars into galaxies.

Key idea

You see three groups at a glance. K-means finds them — in any number of dimensions.

K-means, Step by Step

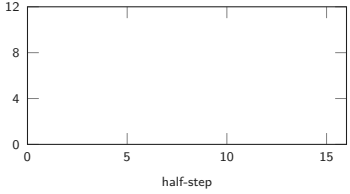
Watch K-means work ($K = 3$)



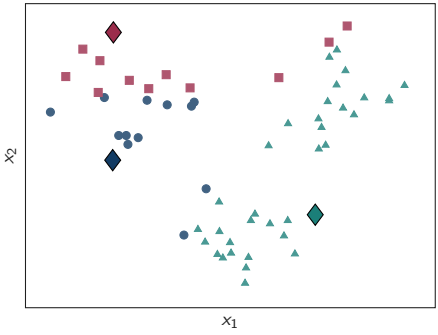
Step

Initialise. Pick K training examples at random as the first centroids μ_k (diamonds).

distortion J after each half-step



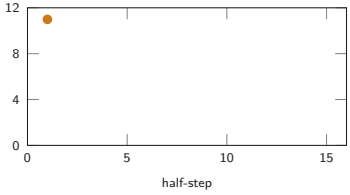
Watch K-means work ($K = 3$)



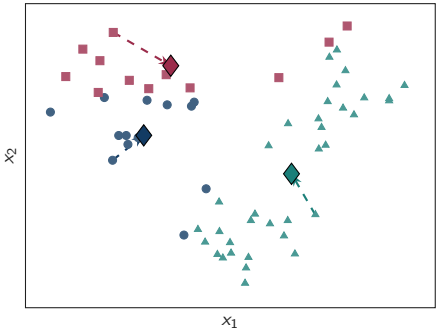
Step

Iteration 1, assign. Each point joins its nearest centroid. $J = 10.990$.

distortion J after each half-step

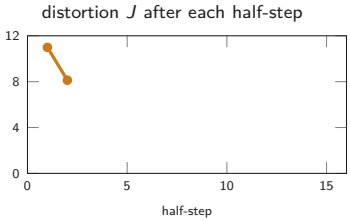


Watch K-means work ($K = 3$)

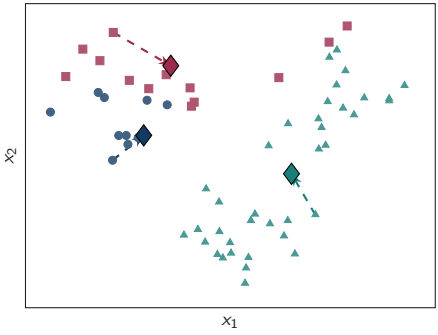


Step

Iteration 1, move. Each centroid moves to the mean of its points. $J = 8.124$.



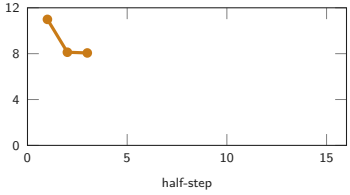
Watch K-means work ($K = 3$)



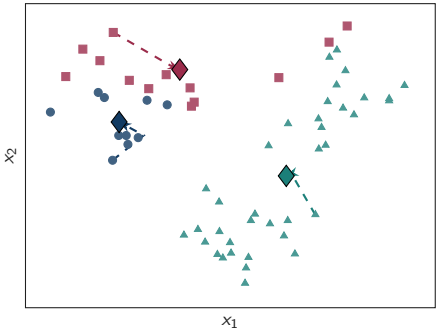
Step

Iteration 2, assign. Each point joins its nearest centroid. $J = 8.062$.

distortion J after each half-step



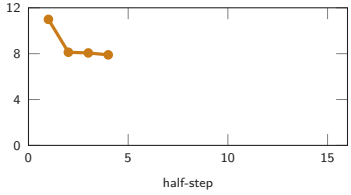
Watch K-means work ($K = 3$)



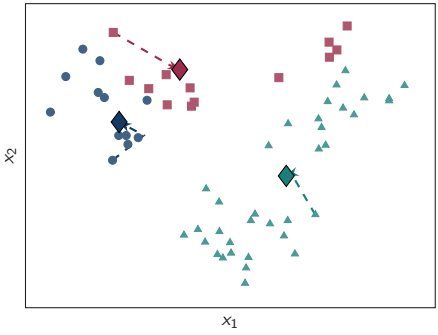
Step

Iteration 2, move. Each centroid moves to the mean of its points. $J = 7.896$.

distortion J after each half-step



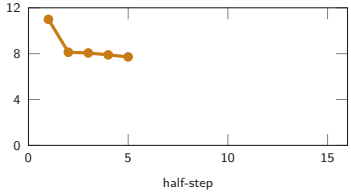
Watch K-means work ($K = 3$)



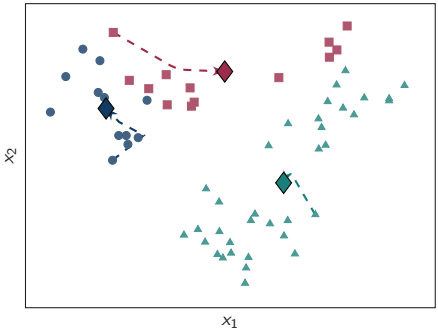
Step

Iteration 3, assign. Each point joins its nearest centroid. $J = 7.716$.

distortion J after each half-step



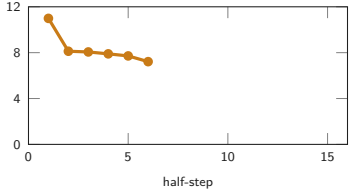
Watch K-means work ($K = 3$)



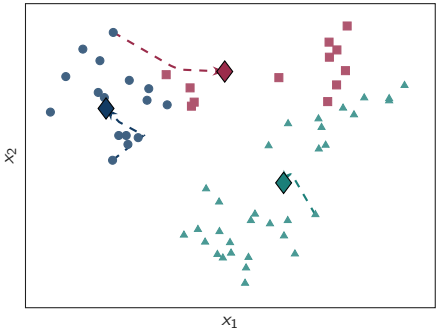
Step

Iteration 3, move. Each centroid moves to the mean of its points. $J = 7.216$.

distortion J after each half-step



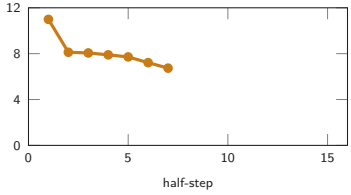
Watch K-means work ($K = 3$)



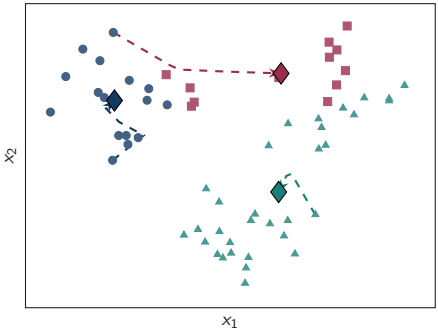
Step

Iteration 4, assign. Each point joins its nearest centroid. $J = 6.737$.

distortion J after each half-step

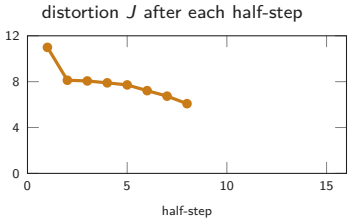


Watch K-means work ($K = 3$)

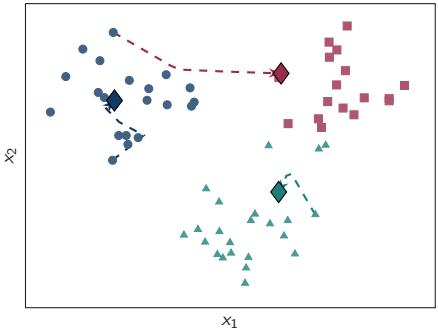


Step

Iteration 4, move. Each centroid moves to the mean of its points. $J = 6.080$.



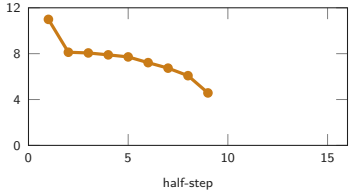
Watch K-means work ($K = 3$)



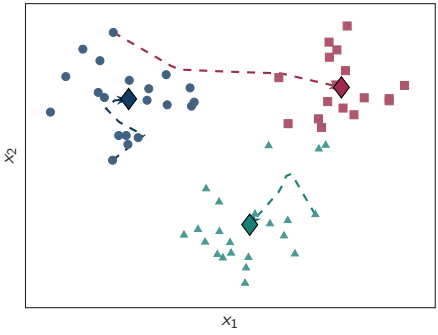
Step

Iteration 5, assign. Each point joins its nearest centroid. $J = 4.575$.

distortion J after each half-step



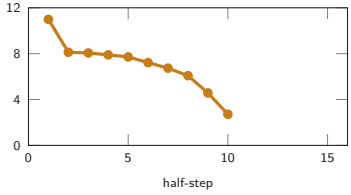
Watch K-means work ($K = 3$)



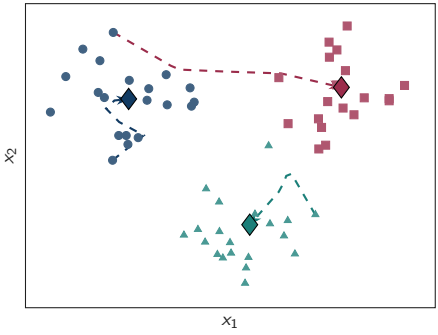
Step

Iteration 5, move. Each centroid moves to the mean of its points. $J = 2.714$.

distortion J after each half-step



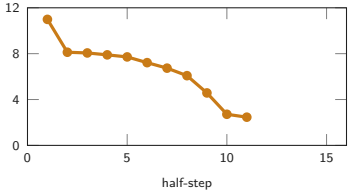
Watch K-means work ($K = 3$)



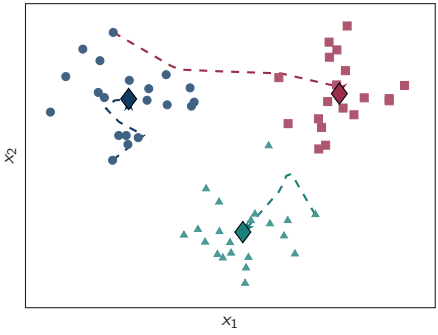
Step

Iteration 6, assign. Each point joins its nearest centroid. $J = 2.460$.

distortion J after each half-step



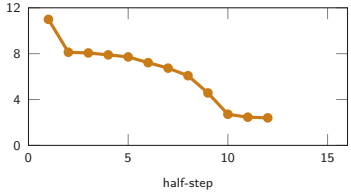
Watch K-means work ($K = 3$)



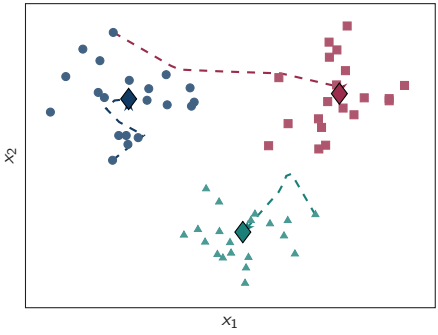
Step

Iteration 6, move. Each centroid moves to the mean of its points. $J = 2.401$.

distortion J after each half-step



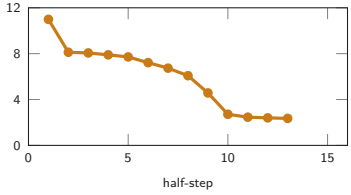
Watch K-means work ($K = 3$)



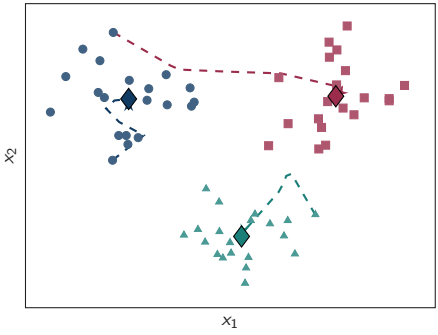
Step

Iteration 7, assign. Each point joins its nearest centroid. $J = 2.354$.

distortion J after each half-step



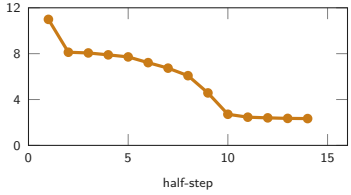
Watch K-means work ($K = 3$)



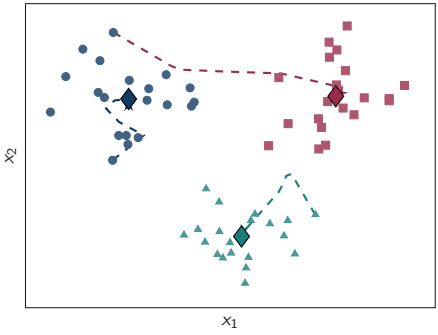
Step

Iteration 7, move. Each centroid moves to the mean of its points. $J = 2.338$.

distortion J after each half-step



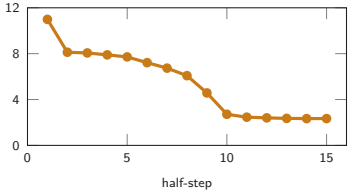
Watch K-means work ($K = 3$)



Step

Iteration 8, assign. No point changes cluster: converged. $J = 2.338$.

distortion J after each half-step



What you just watched

Two steps, repeated

1. **Assign** every point to its nearest centroid (recolour).
2. **Move** every centroid to the mean of the points assigned to it.

Stop when an assignment step changes nothing.

This run started badly. Two of the three random starting points (blue and rose) lay in the top-left group, so teal had to cover both other groups — 35 of the 60 points. Rose was then pulled across the top one move at a time, its x_1 going $1.6 \to 3.3 \to 3.5 \to 4.8 \to 6.5 \to 8.3$, until it reached the top-right group. Teal was left with the bottom group, and the clusters settled at 20/20/20.

J never goes up

The cost fell at every half-step: from 10.99 after the first assignment to 2.338 at convergence. That is no accident — it is guaranteed, and the next section but one proves it.

The numbers

Iterations: 8. Final centroids: $(2.02, 7.76)$, $(8.09, 7.86)$, $(5.33, 2.65)$.

The K-means Algorithm

K-means, precisely

K-means

Randomly initialise K cluster centroids $\mu_1, \ldots, \mu_K \in \mathbb{R}^n$.

Repeat {

  for $i = 1$ to m:

    $c^{(i)} :=$ index k of the centroid closest to $\mathbf{x}^{(i)}$,

    i.e. $c^{(i)} = \arg\min_k \|\mathbf{x}^{(i)} - \mu_k\|^2$

  for $k = 1$ to K:

    $\mu_k :=$ mean of the points assigned to cluster k

}

Notation

$c^{(i)} \in \{1, \ldots, K\}$: the cluster of example i.

$\mu_k \in \mathbb{R}^n$: the centroid of cluster k — a point in feature space, not a training example.

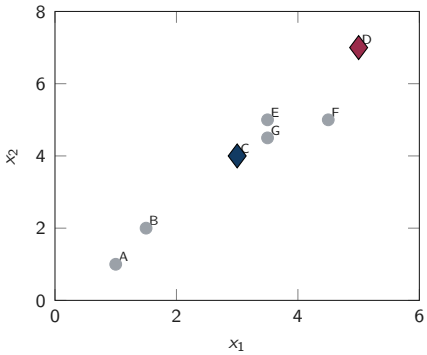
$\mu_{c^{(i)}}$: the centroid of the cluster that $\mathbf{x}^{(i)}$ is in.

An empty cluster

If no point is assigned to some μ_k, its mean is $0/0$. Either drop that cluster (continue with $K - 1$) or re-initialise it at a random example. In our data, starting all three centroids bunched in one corner produces exactly this.

Squared distance is used because it is cheaper (no square root) and selects the same nearest centroid.

By hand: seven points, $K = 2$



Data

A$(1, 1)$, B$(1.5, 2)$, C$(3, 4)$, D$(5, 7)$, E$(3.5, 5)$, F$(4.5, 5)$, G$(3.5, 4.5)$. Start: $\mu_1 = \text{C}$, $\mu_2 = \text{D}$.

Step

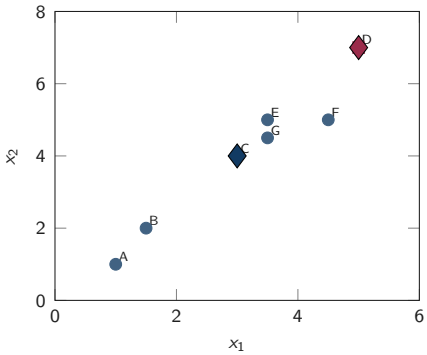
Initialise. Pick K training examples at random as the first centroids μ_k (diamonds).

$\mu_1 = (3,\ 4)$, $\mu_2 = (5,\ 7)$

Check one

First assignment, point F: $\|F - \mu_1\|^2 = 1.5^2 + 1^2 = 3.25$, $\|F - \mu_2\|^2 = 0.5^2 + 2^2 = 4.25$. F goes to cluster 1.

By hand: seven points, $K = 2$



Data

A$(1, 1)$, B$(1.5, 2)$, C$(3, 4)$, D$(5, 7)$, E$(3.5, 5)$, F$(4.5, 5)$, G$(3.5, 4.5)$. Start: $\mu_1 = $ C, $\mu_2 = $ D.

Step

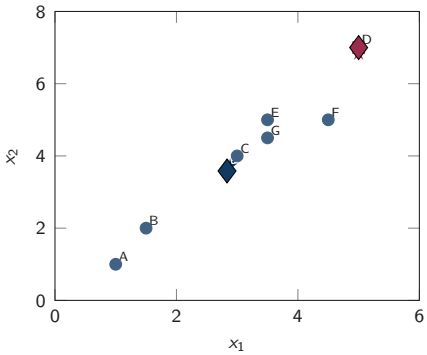
Iteration 1, assign. Each point joins its nearest centroid. $J = 3.464$.

$\mu_1 = (3,\ 4)$, $\mu_2 = (5,\ 7)$

Check one

First assignment, point F: $\|F - \mu_1\|^2 = 1.5^2 + 1^2 = 3.25$, $\|F - \mu_2\|^2 = 0.5^2 + 2^2 = 4.25$. F goes to cluster 1.

By hand: seven points, $K = 2$



Data

A$(1, 1)$, B$(1.5, 2)$, C$(3, 4)$, D$(5, 7)$, E$(3.5, 5)$, F$(4.5, 5)$, G$(3.5, 4.5)$. Start: $\mu_1 = $ C, $\mu_2 = $ D.

Step

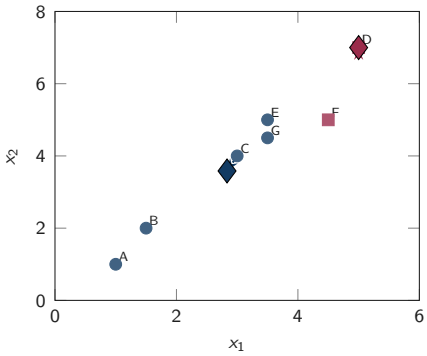
Iteration 1, move. Each centroid moves to the mean of its points. $J = 3.292$.

$\mu_1 = (2.83,\ 3.58)$, $\mu_2 = (5,\ 7)$

Check one

First assignment, point F: $\|F - \mu_1\|^2 = 1.5^2 + 1^2 = 3.25$, $\|F - \mu_2\|^2 = 0.5^2 + 2^2 = 4.25$. F goes to cluster 1.

By hand: seven points, $K = 2$



Data

A$(1,1)$, B$(1.5,2)$, C$(3,4)$, D$(5,7)$, E$(3.5,5)$, F$(4.5,5)$, G$(3.5,4.5)$. Start: $\mu_1 =$ C, $\mu_2 =$ D.

Step

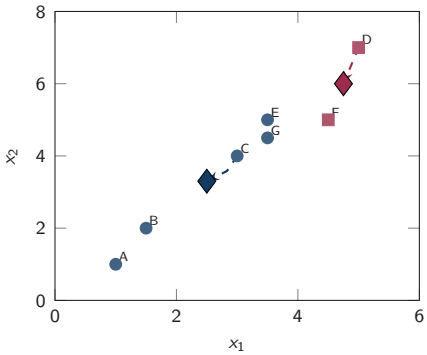
Iteration 2, assign. Each point joins its nearest centroid. $J = 3.215$.

$\mu_1 = (2.83,\ 3.58)$, $\mu_2 = (5,\ 7)$

Check one

First assignment, point F: $\|F - \mu_1\|^2 = 1.5^2 + 1^2 = 3.25$, $\|F - \mu_2\|^2 = 0.5^2 + 2^2 = 4.25$. F goes to cluster 1.

By hand: seven points, $K = 2$



Data

A$(1, 1)$, B$(1.5, 2)$, C$(3, 4)$, D$(5, 7)$, E$(3.5, 5)$, F$(4.5, 5)$, G$(3.5, 4.5)$. Start: $\mu_1 = $ C, $\mu_2 = $ D.

Step

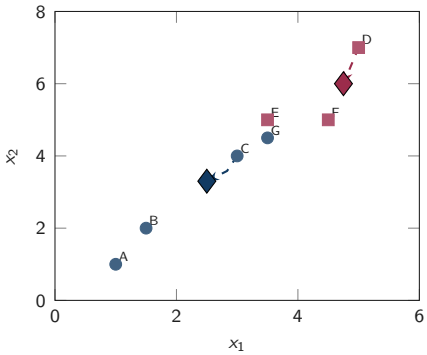
Iteration 2, move. Each centroid moves to the mean of its points. $J = 2.775$.

$\mu_1 = (2.5,\ 3.3)$, $\mu_2 = (4.75,\ 6)$

Check one

First assignment, point F: $\|F - \mu_1\|^2 = 1.5^2 + 1^2 = 3.25$, $\|F - \mu_2\|^2 = 0.5^2 + 2^2 = 4.25$. F goes to cluster 1.

By hand: seven points, $K = 2$



Data

A$(1,1)$, B$(1.5,2)$, C$(3,4)$, D$(5,7)$, E$(3.5,5)$, F$(4.5,5)$, G$(3.5,4.5)$. Start: $\mu_1 = $ C, $\mu_2 = $ D.

Step

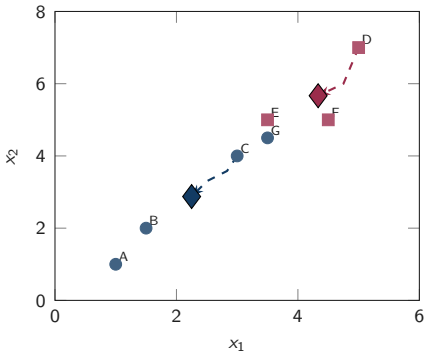
Iteration 3, assign. Each point joins its nearest centroid. $J = 2.585$.

$\mu_1 = (2.5,\ 3.3)$, $\mu_2 = (4.75,\ 6)$

Check one

First assignment, point F: $\|F - \mu_1\|^2 = 1.5^2 + 1^2 = 3.25$, $\|F - \mu_2\|^2 = 0.5^2 + 2^2 = 4.25$. F goes to cluster 1.

By hand: seven points, $K = 2$



Data

A$(1, 1)$, B$(1.5, 2)$, C$(3, 4)$, D$(5, 7)$, E$(3.5, 5)$, F$(4.5, 5)$, G$(3.5, 4.5)$. Start: $\mu_1 = $ C, $\mu_2 = $ D.

Step

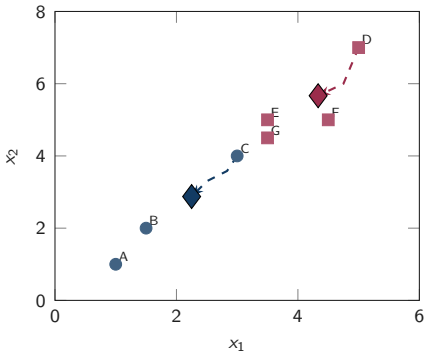
Iteration 3, move. Each centroid moves to the mean of its points. $J = 2.324$.

$\mu_1 = (2.25,\ 2.88)$, $\mu_2 = (4.33,\ 5.67)$

Check one

First assignment, point F: $\|F - \mu_1\|^2 = 1.5^2 + 1^2 = 3.25$, $\|F - \mu_2\|^2 = 0.5^2 + 2^2 = 4.25$. F goes to cluster 1.

By hand: seven points, $K = 2$



Data

A$(1,1)$, B$(1.5,2)$, C$(3,4)$, D$(5,7)$, E$(3.5,5)$, F$(4.5,5)$, G$(3.5,4.5)$. Start: $\mu_1 =$ C, $\mu_2 =$ D.

Step

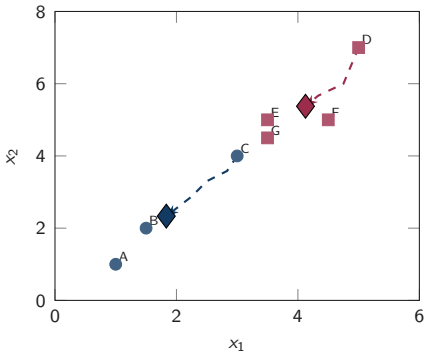
Iteration 4, assign. Each point joins its nearest centroid. $J = 2.018$.

$\mu_1 = (2.25,\ 2.88)$, $\mu_2 = (4.33,\ 5.67)$

Check one

First assignment, point F: $\|F - \mu_1\|^2 = 1.5^2 + 1^2 = 3.25$, $\|F - \mu_2\|^2 = 0.5^2 + 2^2 = 4.25$. F goes to cluster 1.

By hand: seven points, $K = 2$



Data

A$(1, 1)$, B$(1.5, 2)$, C$(3, 4)$, D$(5, 7)$, E$(3.5, 5)$, F$(4.5, 5)$, G$(3.5, 4.5)$. Start: $\mu_1 =$ C, $\mu_2 =$ D.

Step

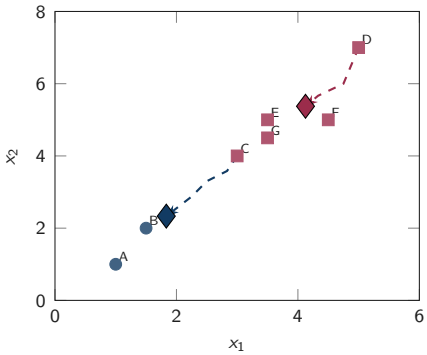
Iteration 4, move. Each centroid moves to the mean of its points. $J = 1.744$.

$\mu_1 = (1.83,\ 2.33)$, $\mu_2 = (4.12,\ 5.38)$

Check one

First assignment, point F: $\|F - \mu_1\|^2 = 1.5^2 + 1^2 = 3.25$, $\|F - \mu_2\|^2 = 0.5^2 + 2^2 = 4.25$. F goes to cluster 1.

By hand: seven points, $K = 2$



Data

A$(1, 1)$, B$(1.5, 2)$, C$(3, 4)$, D$(5, 7)$, E$(3.5, 5)$, F$(4.5, 5)$, G$(3.5, 4.5)$. Start: $\mu_1 = $ C, $\mu_2 = $ D.

Step

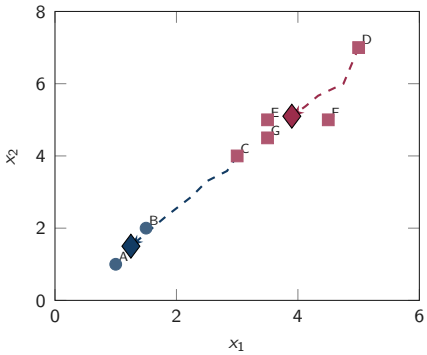
Iteration 5, assign. Each point joins its nearest centroid. $J = 1.604$.

$\mu_1 = (1.83,\ 2.33)$, $\mu_2 = (4.12,\ 5.38)$

Check one

First assignment, point F: $\|F - \mu_1\|^2 = 1.5^2 + 1^2 = 3.25$, $\|F - \mu_2\|^2 = 0.5^2 + 2^2 = 4.25$. F goes to cluster 1.

By hand: seven points, $K = 2$



Data

A$(1, 1)$, B$(1.5, 2)$, C$(3, 4)$, D$(5, 7)$, E$(3.5, 5)$, F$(4.5, 5)$, G$(3.5, 4.5)$. Start: $\mu_1 =$ C, $\mu_2 =$ D.

Step

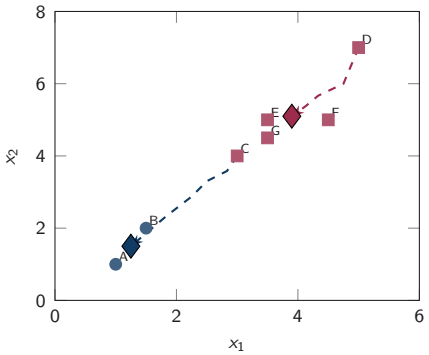
Iteration 5, move. Each centroid moves to the mean of its points. $J = 1.218$.

$\mu_1 = (1.25,\ 1.5)$, $\mu_2 = (3.9,\ 5.1)$

Check one

First assignment, point F: $\|F - \mu_1\|^2 = 1.5^2 + 1^2 = 3.25$, $\|F - \mu_2\|^2 = 0.5^2 + 2^2 = 4.25$. F goes to cluster 1.

By hand: seven points, $K = 2$



Data

A$(1,1)$, B$(1.5,2)$, C$(3,4)$, D$(5,7)$, E$(3.5,5)$, F$(4.5,5)$, G$(3.5,4.5)$. Start: $\mu_1 =$ C, $\mu_2 =$ D.

Step

Iteration 6, assign. No point changes cluster: converged. $J = 1.218$.

$\mu_1 = (1.25,\ 1.5)$, $\mu_2 = (3.9,\ 5.1)$

Check one

First assignment, point F: $\|F - \mu_1\|^2 = 1.5^2 + 1^2 = 3.25$, $\|F - \mu_2\|^2 = 0.5^2 + 2^2 = 4.25$. F goes to cluster 1.

The Optimisation Objective

K-means minimises the distortion

Distortion (cost function)

$$J\left(c^{(1)}, \ldots, c^{(m)}, \mu_1, \ldots, \mu_K\right) = \frac{1}{m} \sum_{i=1}^{m} \left\| \mathbf{x}^{(i)} - \mu_{c^{(i)}} \right\|^2$$

The average squared distance from each point to the centroid of its cluster.

K-means is **coordinate descent** on J: it minimises over one group of variables while holding the other fixed, and alternates.

Why neither step can increase J

Assign (hold μ fixed, choose c): each term $\|\mathbf{x}^{(i)} - \mu_{c^{(i)}}\|^2$ is made as small as possible by the nearest centroid.

Move (hold c fixed, choose μ_k): $\sum_{i:c^{(i)}=k} \|\mathbf{x}^{(i)} - \mu_k\|^2$ is minimised by the mean. Setting its gradient $-2\sum_{i:c^{(i)}=k}(\mathbf{x}^{(i)} - \mu_k)$ to zero gives exactly $\mu_k =$ mean.

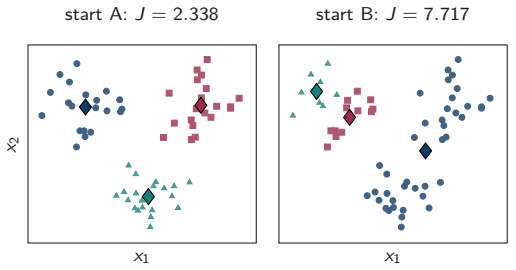
Consequences

J decreases or stays the same at every step, and there are finitely many assignments, so K-means always converges. If J ever rises, the code has a bug.

Hand example: $J = 3.464 \to 3.292 \to \cdots \to 1.218$.

Initialising K-means

Convergence is guaranteed — to a *local* minimum



Same data, same algorithm, different random starting centroids.

Random initialisation

Choose $K < m$. Pick K training examples at random; set $\mu_1, \ldots, \mu_K$ to them.

200 random starts on these 60 points

193 reach the best clustering, $J = 2.338$.
7 stop in a poor one, J between 7.717 and 8.152: one centroid covers two groups while two split the third.

Multiple random initialisations

Recipe

for $t = 1$ to 100 (typically 50–1000):
  randomly initialise K-means;
  run it to convergence, giving $c^{(i)}$, μ_k;
  compute the distortion J.
Keep the run with the **lowest** J.

How many runs are enough?

If one run finds the best clustering with probability p, then t runs all miss it with probability $(1-p)^t$. Here $p = 193/200 = 0.965$: two runs miss with probability $0.035^2 \approx 0.0012$.

Key idea

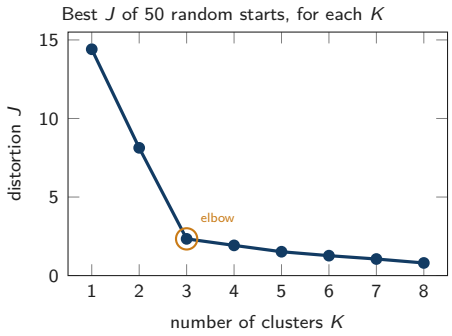
Comparing runs needs a number, and J is that number — one more use of having an explicit objective.

Harder problems

With many clusters or high dimensions p can be small; that is when hundreds of restarts, or smarter seeding such as k-means++, pay off.

Choosing the Number of Clusters

The elbow method



Reading the curve

J falls steeply up to $K = 3$ ($14.41 \to 8.13 \to 2.34$) and slowly after ($1.92, 1.52, \ldots$). The bend — the "elbow" — suggests $K = 3$.

Do not choose K by minimising J

J always decreases as K grows ($K = m$ gives $J = 0$). And many real data sets have no clear elbow: the curve bends smoothly.

Choose K for the purpose

T-shirt sizes

Cluster customers by height and weight. $K = 3$ gives S, M, L; $K = 5$ gives XS, S, M, L, XL. Neither is "correct". Five sizes fit customers better; three are cheaper to manufacture and stock. The business decides.

Key idea

Clustering is usually a step towards something else. Evaluate K by how well the clusters serve that later purpose.

Other practical uses

- **Image compression**: cluster pixel colours into $K = 16$ colours and store one 4-bit index per pixel instead of 24 bits.
- **Pre-processing**: use cluster membership as a feature for a supervised model.

Questions to ask

How many groups can the user act on? What does one more cluster cost? Do the clusters stay stable across random restarts?

Wrap-up and Assessment

Summary

What to remember

- Clustering finds groups in unlabelled data.
- K-means alternates: assign each point to its nearest centroid; move each centroid to its cluster's mean.
- It is coordinate descent on the distortion J, which never increases.
- It converges, but possibly to a local minimum: run many random initialisations and keep the lowest J.
- Choose K by the elbow if there is one — and by the downstream purpose in any case.

Formulas

$$c^{(i)} = \arg\min_k \|\mathbf{x}^{(i)} - \mu_k\|^2$$

$$\mu_k = \frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{x}^{(i)}$$

$$J = \frac{1}{m} \sum_i \|\mathbf{x}^{(i)} - \mu_{c^{(i)}}\|^2$$

Next

Part 2: anomaly detection — the Gaussian distribution, density estimation, and how to evaluate a detector with very few anomalies.

Descriptive questions

1. Distinguish supervised from unsupervised learning, and give three applications of clustering. **[4]**
2. State the K-means algorithm. Explain the meaning of $c^{(i)}$, μ_k and $\mu_{c^{(i)}}$. **[5]**
3. Run K-means by hand on the seven points of the lecture, $K = 2$, starting from $\mu_1 = A$, $\mu_2 = D$. Give the assignments and centroids at each iteration. **[8]**
4. Define the distortion J. Prove that neither step of K-means can increase it. **[8]**
5. What happens when a cluster receives no points, and what are two remedies? **[4]**
6. Why can K-means converge to different solutions from different starts? Describe random initialisation with multiple restarts. **[5]**

Descriptive questions (continued)

7. A single random start finds the best clustering with probability 0.3. How many starts are needed for at least a 99% chance of finding it? **[5]**
8. Explain the elbow method and two reasons it can fail. **[5]**
9. Why does J always decrease as K increases (with the best initialisation)? What is J when $K = m$? **[4]**
10. Explain how K-means compresses an image to 16 colours, and compute the compression factor for a 256×256 RGB image, ignoring the colour table. **[6]**
11. Give an example where the right K is set by the application rather than by the data. **[4]**
12. Explain why K-means uses squared distance in the assignment step, and why the mean is the right centroid for that choice. **[5]**

Design questions

1. **Design a segmentation.** A Chattogram supermarket chain has a year of purchase records for 200 000 loyalty-card holders. Design a K-means segmentation: the features (and their scaling), how you choose K, and how marketing will use the clusters. **[12]**
2. **Design for scale.** K-means on 10^7 points with $K = 100$ is too slow to restart 500 times. Design a cheaper procedure that still guards against poor local minima, and say how you would check it. **[10]**
3. **Design a test of a K-means implementation.** Design three small data sets with known answers that would expose: a wrong assignment rule, a wrong centroid update, and missing empty-cluster handling. **[10]**
4. **Design a stability check.** Design an experiment that tells a client whether the clusters you report are real structure or an artefact of the random initialisation. **[12]**

Reading

Primary text

G. James, D. Witten, T. Hastie and R. Tibshirani, *An Introduction to Statistical Learning*, 2nd ed., Springer, 2021.

→ **§12.1** — the challenge of unsupervised learning; **§12.4.1** — K-means clustering, its objective, and local optima.

Supplementary

C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.

→ **§9.1** — K-means clustering as alternating minimisation of the distortion measure, and image compression.

Lab (CSE 816, Week 7): K-means.